The empirical recurrence for OEIS A233923: recovering a conjecture that is not written down

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Abstract

OEIS A233923 records "Empirical recurrence of order 82", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 625$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233923 is "Number of (n+1)X(3+1) 0..4 arrays with every 2X2 subblock having the sum of the absolute values of the edge differences equal to 8 and no adjacent elements equal.". It has offset 1 and begins

$$2792, 32496, 405004, 5389600, 73564176, 1020495508, 14257498880, 200008938060, \ \ldots$$

The entry states:

> Empirical recurrence of order 82 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:15:56, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 625$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1250$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{82} c_i \, a(n-i),$$

c_{1}	30	c_{22}	358871370356739006	c_{43}	−5305811028948073180705546	c_{64}	317
c_{2}	−41	c_{23}	5335817120797912981	c_{44}	−7462353560274124158529736	c_{65}	745
c_{3}	−6745	c_{24}	−7207048326212368387	c_{45}	10073352433811188668406413	c_{66}	−49
c_{4}	50230	c_{25}	−43440874456292182642	c_{46}	8803599204036324786067402	c_{67}	−5
c_{5}	595825	c_{26}	85715771675285356183	c_{47}	−15098526186874086346480987	c_{68}	51
c_{6}	−7351028	c_{27}	278621562459493458601	c_{48}	−7862303360798692296860116	c_{69}	17
c_{7}	−22832572	c_{28}	−741843426077845664011	c_{49}	18003997668817654980386900	c_{70}	−3
c_{8}	551655707	c_{29}	−1387769655441171569180	c_{50}	4859388509825444373228894	c_{71}	5
c_{9}	−106308546	c_{30}	4973267250801408723962	c_{51}	−17089515174733679765225895	c_{72}	1
c_{10}	−26004408747	c_{31}	5147032653162817911531	c_{52}	−1419634200797792232325849	c_{73}	−
c_{11}	55317448700	c_{32}	−26543887870125421781248	c_{53}	12859010878940902643953247	c_{74}	−
c_{12}	832721839752	c_{33}	−12387911916876612162104	c_{54}	−790138368983708929833251	c_{75}	
c_{13}	−3062334448054	c_{34}	114359337253735234030819	c_{55}	−7607343933842006300442854	c_{76}	
c_{14}	−18650697157606	c_{35}	5446216078314120915176	c_{56}	1348254923704510937288636	c_{77}	
c_{15}	100138235881038	c_{36}	−400409387433490190107998	c_{57}	3494404450787498087210317	c_{78}	
c_{16}	287805372584569	c_{37}	113349605209987574776488	c_{58}	−956885720218972794883159	c_{79}	
c_{17}	−2294844217991054	c_{38}	1141891316334767350625713	c_{59}	−1224220792965398098812628	c_{80}	
c_{18}	−2707972919013118	c_{39}	−641732540989799573708502	c_{60}	439671591973365789666671	c_{81}	
c_{19}	39253544546232183	c_{40}	−2648380729996730228078784	c_{61}	318854323490414734152075	c_{82}	
c_{20}	3851202417314097	c_{41}	2159646805227968712909462	c_{62}	−140782135853496786826071		
c_{21}	−517124161558145841	c_{42}	4968705180998850471675193	c_{63}	−59441908505652634435834		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82+4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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